

Analytical modeling and experimental validation of a base-isolated statue: the case study of the Goddess of Morgantina

Maria Oliva¹[0000-0002-5118-6699], Laura Floriano¹[0000-0001-6468-6660],
Mariangela Liuzzo¹[0000-0003-3558-8601], Francesco Lo
Iacono¹[0000-0003-1035-8017], Giuseppe Mugnos¹[0009-0009-8805-1526], and
Giacomo Navarra¹[0000-0001-7997-8770]

Department of Engineering and Architecture, Kore University of Enna, Enna, Italy
`maria.oliva@unikore.it`

Abstract. The preservation of cultural heritage artifacts in seismic regions is increasingly important due to their vulnerability during earthquakes. While much attention has been devoted to historic structures, museum contents such as statues remain less protected. To address this gap, a novel bi-directional base isolation system using the patented Kinematic Steel Joint (KSJ) has been developed. The KSJ features a double articulated steel quadrilateral with crossing rods, providing self-centering, energy dissipation, and pendulum-like behavior in a compact and cost-effective design. An extensive experimental campaign was carried out at the L.E.D.A. Research Institute of Kore University of Enna. The testing included quasi-static trials for geometric assessment, broadband noise excitation for evaluating dynamic transmissibility, and shake table simulations based on real earthquake records to assess system performance in terms of transmitted accelerations and displacements. A full-scale replica of the Statue of the Goddess of Morgantina was mounted on the KSJ device to assess its protective efficiency under seismic loading. The isolator effectiveness in reducing seismic accelerations transmitted to the statue had already been demonstrated in previous studies by the same authors. In this study, an analytical model of the coupled KSJ-statue system is developed, calibrated and validated against experimental data, providing an accurate numerical representation of its dynamic response.

Keywords: cultural heritage protection · base-isolation system · analytical modelling · friction pendulum bearing.

1 Introduction

The protection of statues, artifacts and museum collections during seismic events has become an increasingly critical area of research in earthquake engineering and heritage conservation. Cultural heritage items are often composed of brittle, irreplaceable materials, posing unique engineering challenges due to their complex and unpredictable dynamic responses. Recent earthquakes in Mediterranean

regions (e.g., Italy, Greece, Croatia and Turkey) and beyond have highlighted the vulnerability of such items, emphasizing the urgent need for scientifically validated and non-invasive protection strategies.

Among the various approaches, base isolation has emerged as the most widely adopted solution, decoupling the artifact from ground motion and reducing seismic forces. Foundational work by Augusti et al. [1] introduced strategies such as sliding bases, center-of-mass lowering, and isolation supports. Building on classical rocking block theory [2–4], subsequent research refined these methods, exploring isolation devices within museum pedestals [5–7], including friction pendulum systems [8–10] and elastomeric bearings [11, 12]. Experimental studies using shake tables have validated these strategies on display cases [13, 14], pedestal-mounted isolators [15, 16], and scaled artifact models [17, 18]. Recent work by Berto et al. [19] investigated the effectiveness of double-concave curved-surface slider isolators across various artifact geometries.

Building on these foundations, in our previous work [20], we dynamically characterized the Kinematic Steel Joint (KSJ) isolator, a bi-directional base isolation device designed to protect cultural artifacts, and in particular the Goddess of Morgantina, selected as a case study. The KSJ consists of articulated quadrilateral steel mechanisms arranged in orthogonal layers, allowing independent horizontal movement along both axes following a double-curved concave surface [21]. This enables self-centering and energy dissipation through dry friction at pinned joints. Manufactured from folded sheet steel, the device combines structural robustness, modularity, low production costs, and accommodates horizontal displacements of ± 150 mm per direction with integrated shock absorbers. While effective for horizontal seismic mitigation, the KSJ does not provide vertical isolation, similar to conventional systems.

The KSJ’s dynamic behavior was extensively investigated at the L.E.D.A. Research Institute of Kore University of Enna using a six-degrees-of-freedom shaking table [22]. Tests included quasi-static loading, broadband excitations, and seismic simulations based on historical earthquake records. Results demonstrated an elliptical paraboloid platform motion with nearly equal principal curvatures along x and y axes, significant attenuation of high-frequency vibrations, stable payload-independent dynamic responses, and excellent re-centering with residual displacements below 20 mm [20]. These findings validate the KSJ isolator as a reliable passive system for seismic protection of cultural artifacts. Furthermore, to assess the effectiveness of the proposed base isolation system in protecting statues, dynamic tests were conducted on a full-scale replica of the Goddess of Morgantina, as testing the actual artifact was considered impractical.

In this paper, an analytical model of the device is developed and calibrated based on the results of a full-scale shaking table test on the KSJ isolator, which was used to protect a full-scale replica of the Goddess of Morgantina, as previously mentioned. Unlike the approach adopted in Ref. [23], where the statue’s mass was considered concentrated at the base isolation level and assumed rigidly connected to it, the statue’s displacements in this study were derived by assuming that the entire mass is concentrated at the centroid, taking into account the

rotational displacements of the statue with respect to the isolated base, which were already observed during the dynamic tests. It is important to note that overturning is not considered in this case because the replica of the Goddess of Morgantina was constructed to replicate the original statue's characteristics. The original is composed of three blocks connected by bars, which are also anchored to the existing uni-directional base isolator, thereby preventing overturning. Therefore, in this case, the rotational motion of the statue about the base, constrained by two bars that prevent detachment or overturning, can be modeled using a rotational stiffness.



Fig. 1. (a) The statue of the Goddess of Morgantina at Aidone's Archaeological Museum; (b) the full-scale replica with the isolation device on the shaking tables at L.E.D.A. [20].

2 Digital modeling and replication of the Goddess of Morgantina

The statue of the Goddess of Morgantina, currently exhibited at the Archaeological Museum of Aidone (Figure 1a), was selected as a case study for its Sicilian origin and complex history [24, 25]. Dating back to the 5th century BC, the 2.24 m-high sculpture, made of marble and calcareous tuff, has become a symbol of the fight against the illicit trafficking of cultural heritage. Illegally excavated in

the late 1970s, it was divided into three parts to ease export, sold in Switzerland in 1986, and later acquired by the J. Paul Getty Museum in Malibu. After a long legal dispute, it was repatriated to Italy in 2011 together with a unidirectional base isolator originally designed for its protection.

Since seismic testing on the original artifact was not possible, a full-scale replica of the Goddess of Morgantina was produced, matching the original in both dimensions and mass. The Laboratory of Survey and Representation at Kore University of Enna created a high-resolution 3D model through combined laser scanning and photo-modelling, ensuring an accurate reproduction of the geometry and dynamic properties [26]. From the point cloud, a textured mesh was generated to extract key parameters reported in Table 1 and to optimize the formwork subdivision.

Table 1. Main characteristics of the Morgantina Goddess statue

Parameter	Value
Mass m_s	908 kg
Volume V_s	0.356 m ³
Center of Gravity Height h	0.979 m
Moments of Inertia $I_{0,x}$	0.0968 m ⁴
Moments of Inertia $I_{0,y}$	0.1013 m ⁴

The replica was cast using a fiber-reinforced cementitious mortar, whose density matched the original material. The formwork was additively manufactured in PLA using Fused Deposition Modeling (FDM), divided into 45 cm segments, which were cast and assembled to reconstruct the statue (Figure 1b). The blocks were connected by two threaded steel rods, replicating the reinforcement system applied to the original after its return to Sicily. Further details on the geometric survey and the manufacturing of the statue are reported in Refs. [20, 26].

3 Experimental setup

The replica was mounted on the KSJ isolator and subjected to bi-directional seismic tests using the Kobe earthquake record (CUE, Takatori station, Japan, 16/01/1995, $M_w = 6.9$, $PGA_x = 0.641$ g, $PGA_y = 0.439$ g). The tests were carried out on the shaking table of the L.E.D.A. Research Institute at the University Kore of Enna, with input motion scaled from 20% to 120% of its original amplitude to assess the system’s response under increasing seismic demand. The statue–isolator system was instrumented with a comprehensive monitoring setup to capture its seismic response. Two tri-axial piezoelectric high-sensitivity accelerometers were installed to record accelerations at the center of the shaking table (R1) and at the upper platform of the isolator (A1), as shown in Figure 1b. As stated in [20], the system proved capable of mitigating seismic accelerations transmitted to the protected artifact.

In addition, a motion capture system with high-resolution infrared cameras and retro-reflective spherical markers was used to obtain absolute displacement time histories and spatial components. The setup included three markers on the shaking table to define the reference system, one marker on the KSJ isolator (D1), and one placed just below the statue head (D2), as illustrated in Figure 1b.

In the present study, particular attention is given to the displacement response of the system, as it provides crucial information for the calibration of the proposed analytical model. In order to achieve this objective, the displacement records obtained from the bi-directional seismic tests at the 100% input level (i.e. corresponding to the full amplitude of the scaled earthquake motion) are analyzed in the following sections.

4 Analytical modelling

Due to the kinematic characteristics of KSJ elements, the base-isolation device under study can be classified as a friction pendulum bearing, since it induces vertical upward displacements as the support moves away from the central rest position. Consequently, the protected item is constrained to follow trajectories contained within a double-curved concave surface.

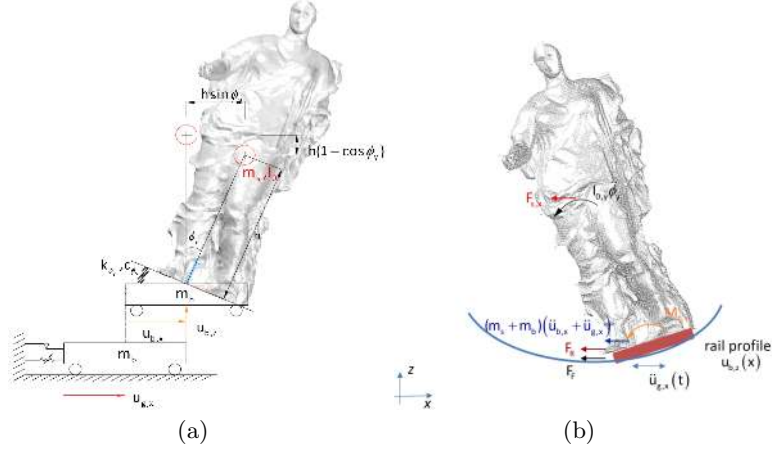


Fig. 2. The considered model: (a) kinematics; (b) mechanics.

The KSJ isolator has been idealized as a mass moving on a bi-directional curved surface, mathematically represented by an elliptical paraboloid, expressed as [20]:

$$u_{b,z} = a_x u_{b,x}^2 + a_y u_{b,y}^2 \quad (1)$$

where a_x and a_y are geometric parameters governing the curvature of the quadric surface; $u_{b,x}$, $u_{b,y}$ and $u_{b,z}$ denote the displacements of the upper platform of the isolator along the three spatial directions.

The statue is connected to the moving mass through a rotational stiffness, which models the mechanical effect of the threaded rods anchoring the statue to the isolator base. These rods allow limited rigid rotation about the base while preventing overturning. Figure 2a shows the kinematics of the system along the x direction, which are equally valid for the y direction. The horizontal and vertical components of the absolute displacement of the center of mass of the statue, u_s , are:

$$\begin{cases} u_{s,x} = u_{g,x} + u_{b,x} + h \sin \phi_y \\ u_{s,y} = u_{g,y} + u_{b,y} + h \sin \phi_x \\ u_{s,z} = u_{b,z} - h(1 - \cos \phi_y) - h(1 - \cos \phi_x) \end{cases} \quad (2)$$

where $u_{g,x}$ and $u_{g,y}$ are the ground displacements in the horizontal directions, h is the vertical distance between the base and the statue's center of mass, ϕ_x and ϕ_y are the small rotations of the statue around the horizontal axes.

The equations of motion governing the statue-isolator assembly consists of two translational and two rotational equations, all referred to the base of the statue. In compact state-space form, the dynamics can be expressed as:

$$\dot{\mathbf{Z}} = \mathbf{D}\mathbf{Z} - \mathbf{G}(\mathbf{Z}, \dot{\mathbf{Z}}) - \tau \ddot{\mathbf{u}}_g^{exp}(t) \quad (3)$$

where the state vector is defined as:

$$\mathbf{Z}^T = [u_{b,x} \quad u_{b,y} \quad \phi_x \quad \phi_y \quad \dot{u}_{b,x} \quad \dot{u}_{b,y} \quad \dot{\phi}_x \quad \dot{\phi}_y] \quad (4)$$

The system matrix $\mathbf{D}$ contains the linear contributions of mass, stiffness, and damping. $\ddot{\mathbf{u}}_g^{exp}$ is the vector capturing the experimental acceleration time histories recorded at the base of the structure during the shaking table seismic tests in the x and y direction, whereas τ is the load location vector. The corresponding definitions are:

$$\mathbf{D} = \begin{bmatrix} \mathbf{0}_{(4 \times 4)} & \mathbf{I}_{(4 \times 4)} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \quad \ddot{\mathbf{u}}_g^{exp}(t) = \begin{bmatrix} \ddot{u}_{g,x}(t) \\ \ddot{u}_{g,y}(t) \end{bmatrix} \quad \tau = \begin{bmatrix} \mathbf{0}_{(4 \times 2)} \\ \mathbf{I}_{(2 \times 2)} \\ \mathbf{0}_{(2 \times 2)} \end{bmatrix} \quad (5)$$

The mass and stiffness matrices are given by:

$$\mathbf{M} = \begin{bmatrix} m_s + m_b & 0 & 0 & 0 \\ 0 & m_s + m_b & 0 & 0 \\ 0 & 0 & I_{0,x} + m_s h^2 & 0 \\ 0 & 0 & 0 & I_{0,y} + m_s h^2 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & k_{\phi_x} & 0 \\ 0 & 0 & 0 & k_{\phi_y} \end{bmatrix} \quad (6)$$

in which m_b and m_s denote the masses of the base isolator and the statue, respectively; $I_{0,x}$ and $I_{0,y}$ are the moments of inertia of the statue about the horizontal axes; k_{ϕ_x} and k_{ϕ_y} represent the rotational stiffnesses associated with

the connection between the statue and the isolator, accounting for both the stiffness of the internal rods and the deformation of the base isolator platform. These allow limited rigid-body rotations while effectively preventing overturning, reproducing the actual experimental boundary conditions.

The system damping is assumed to follow a Rayleigh model:

$$\mathbf{C} = \alpha_0 \mathbf{M} + \alpha_1 \mathbf{K} \quad (7)$$

The damping associated with the rotational motion of the statue is not defined as an independent parameter, as its contribution is considered implicitly included in the global Rayleigh damping model (Eq. 7), which governs both translational and rotational energy dissipation.

Finally, the nonlinear vector $\mathbf{G}(\mathbf{Z}, \dot{\mathbf{Z}})$ collects the restoring forces from the curved surface, the frictional effects, and the additional forces and moments acting on the statue, depicted in Figure 2b for the x direction (the same applies to the y direction):

$$\mathbf{G}(\mathbf{Z}, \dot{\mathbf{Z}}) = \mathbf{A}^{-1} \begin{bmatrix} \mathbf{0}_{(4 \times 1)} \\ (F_{R,x} + F_{F,x} + F_{s,x}) \\ (F_{R,y} + F_{F,y} + F_{s,y}) \\ M_{s,x} \\ M_{s,y} \end{bmatrix} \quad \mathbf{A}^{-1} = \begin{bmatrix} \mathbf{M}^{-1} & \mathbf{0}_{(4 \times 4)} \\ \mathbf{0}_{(4 \times 4)} & \mathbf{M}^{-1} \end{bmatrix} \quad (8)$$

In the base-isolation device, the bi-directional behavior results from the composition of two orthogonal motions at each rail level, depending on a single coordinate. Assuming the surface described by Eq. 1, the horizontal restoring and frictional forces are expressed as [23, 27]:

$$F_{R,i} = (m_s + m_b)g \frac{u'_{b,z(i)}}{1 + u'^2_{b,z(i)}} \quad (i = x, y) \quad (9)$$

$$F_{F,i}(u_{b,i}, \dot{u}_{b,i}) = -(m_s + m_b)g \frac{\mu(\dot{u}_{b,i}) \operatorname{sgn}(\dot{u}_{b,i})}{1 + u'^2_{b,z(i)}} \quad (i = x, y) \quad (10)$$

being g the acceleration due to gravity, $u'_{b,z(i)}$ the slope of the trajectory, $\operatorname{sgn}(\cdot)$ the signum function, $\mu(\dot{u}_{b,i})$ the velocity-variable friction coefficient, defined as [28]:

$$\mu(\dot{u}_{b,i}) = \mu_{\max} - (\mu_{\max} - \mu_{\min}) \exp(-\alpha |\dot{u}_{b,i}|) \quad (i = x, y) \quad (11)$$

where $\mu_{\max}$ and $\mu_{\min}$ are the maximum and minimum mobilized coefficient of friction, respectively, α is a parameter controlling the rate of variation of the friction coefficient and the over dot means time differentiation. Classical friction-based models consider two cases: sliding and sticking. In the sliding phase, the relative velocity is non-zero and the friction force is given by Eq. 10. In the sticking phase, the relative velocity is zero and the friction force is limited by the equilibrium of other tangential forces:

$$F_{F,st}(x, \dot{x}) = -\min(|F_{F,sl}(x, \dot{x})|, |F_{eq}|) \operatorname{sgn}(F_{eq}) \quad (12)$$

where F_{eq} represents the sum of all other forces acting on the protected object in the direction tangential to the trajectory.

In Eq. 8, $F_{s,x}$ and $F_{s,y}$ represent the horizontal inertial forces of the statue, expressed as:

$$F_{s,x} = m_s h (\cos \phi_y \ddot{\phi}_y - \sin \phi_y \dot{\phi}_y^2) \quad (13)$$

$$F_{s,y} = m_s h (\cos \phi_x \ddot{\phi}_x - \sin \phi_x \dot{\phi}_x^2) \quad (14)$$

whereas the moments about the x and y axes are defined as:

$$M_{s,x} = m_s h \ddot{u}_{s,y} \cos \phi_x - m_s h (\ddot{u}_{s,z} - g) \sin \phi_x \quad (15)$$

$$M_{s,y} = m_s h \ddot{u}_{s,x} \cos \phi_y - m_s h (\ddot{u}_{s,z} - g) \sin \phi_y \quad (16)$$

in which $\ddot{u}_{s,i}$ ($i = x, y, z$) are the accelerations of the statue, obtained by differentiating Eq. 2 and expressed as:

$$\begin{cases} \ddot{u}_{s,x} = \ddot{u}_{g,x} + \ddot{u}_{b,x} + h(\cos \phi_y \ddot{\phi}_y - \sin \phi_y \dot{\phi}_y^2) \\ \ddot{u}_{s,y} = \ddot{u}_{g,y} + \ddot{u}_{b,y} + h(\cos \phi_x \ddot{\phi}_x - \sin \phi_x \dot{\phi}_x^2) \\ \ddot{u}_{s,z} = \ddot{u}_{b,z} - h(\cos \phi_y \dot{\phi}_y^2 + \sin \phi_y \ddot{\phi}_y) - h(\cos \phi_x \dot{\phi}_x^2 + \sin \phi_x \ddot{\phi}_x) \end{cases} \quad (17)$$

being $\ddot{u}_{b,i}$ ($i = x, y, z$) the absolute accelerations of the base isolation. The acceleration in z direction, obtained by differentiating Eq.1, is:

$$\ddot{u}_{b,z} = 2a_x (\dot{u}_{b,x}^2 + u_{b,x} \ddot{u}_{b,x}) + 2a_y (\dot{u}_{b,y}^2 + u_{b,y} \ddot{u}_{b,y}) \quad (18)$$

5 Calibration of model parameters

In this study, an optimization procedure has been used to identify the model parameter set that minimizes discrepancies between measured and numerical displacements. The masses of the statue and the base isolation system are assumed as known quantities, whereas the parameters to be optimized include the rotational stiffness of the statue, the Rayleigh damping coefficients, the curvature parameters of the paraboloid defined in Eq. 1 and the frictional properties of the isolators. The resulting parameter vector is defined as:

$$\mathbf{X}^T = [k_{\phi_x} \quad k_{\phi_y} \quad \alpha_0 \quad \alpha_1 \quad a_x \quad a_y \quad \mu_{\max} \quad \mu_{\min} \quad \alpha] \quad (19)$$

To calibrate these parameters, a Particle Swarm Optimization (PSO) algorithm was applied to minimize an objective function defined as the Root Mean Square (RMS) error between the experimental and numerical displacements measured at positions D1 (at the isolator base) and D2 (just below the top of the statue) along both the x and y directions (see Figure 1b):

$$J(\mathbf{X}) = \sum_{j=x,y} \sum_{i=D1,D2} w_{ij} \frac{\text{rms}(u_{i,j}^{\text{num}}(\mathbf{X}, t) - u_{i,j}^{\text{exp}}(t))}{\text{rms}(u_{i,j}^{\text{exp}}(t))}, \quad t \in [t_{\text{in}}, t_f]. \quad (20)$$

where w_{ij} are weighting parameters fixed to 0.25, $\text{rms}(\cdot)$ is the root mean square function, $u_{i,j}$ is the displacement with respect to the ground at i -th position (i.e. D1 and D2) in the j -th global horizontal direction, for the numerical (num) and the experimental (exp) system. The time window (t_i, t_f) have been chosen as the time interval between the 5% and 95% contribution of the experimental shaking table input acceleration time histories $\ddot{\mathbf{u}}_g^{\text{exp}}$ to the Arias intensity [29].

The PSO algorithm updates the parameter vector $\mathbf{X}$ at each iteration, evaluating the system response over the full time interval $[t_i, t_f]$. This is performed by integrating, step by step, the equations of motion of the base-isolated structure with the statue on top, as defined in Eq. 3, using MATLAB's `ode45` solver. At each time step, the relative accelerations of the statue are calculated using Eq. 17, along with the restoring and frictional forces on the isolators defined in Eq. 10-11, while the stick-slip condition is evaluated: if the equilibrium force is less than the available friction, the isolator is assumed to stick and the frictional force is calculated as in Eq 11; otherwise, it slides. Simultaneously, the inertial force and moments of the statue are computed based on Eqs. 13-16. The objective function $J(\mathbf{X})$ in Eq. 20 is then evaluated as the root mean square (RMS) error between the numerical and experimental displacements obtained from the seismic test with the Kobe signal at 100% of its actual intensity.

In order to achieve the optimum value of the parameters, the PSO algorithm has been executed with a swarm of $N_p = 100$ particles and a maximum of $N_i = 200$ iterations, using inertia weight $w = 0.73$ and cognitive and social coefficients $c_1 = c_2 = 1.5$. The optimal parameters, obtained after 46 iterations with $J(\mathbf{X}_{\text{opt}})=0.38$, are:

$$\mathbf{X}_{\text{opt}} = \begin{bmatrix} k_{\phi_x} \\ k_{\phi_y} \\ \alpha_0 \\ \alpha_1 \\ a_x \\ a_y \\ \mu_{\max} \\ \mu_{\min} \\ \alpha \end{bmatrix} = \begin{bmatrix} 3.11 \cdot 10^4 & \text{Nm/rad} \\ 2.57 \cdot 10^4 & \text{Nm/rad} \\ 0.001 \\ 0.839 \\ 0.3070 & \text{m}^{-1} \\ 0.3693 & \text{m}^{-1} \\ 0.016 \\ 0.480 \\ 12.42 \end{bmatrix} \quad (21)$$

Figures 3 and 4 depict the displacement versus time of the base isolator and statue, respectively, in the x and y direction considering the experimental data (red line) and optimal parameters (blue line). In both horizontal directions, the experimental and optimized displacements of the isolator base as well as those of the statue, show very good agreement, especially during the most intense part of the motion, where the largest amplitudes occur.

It is noted that the displacements of the statue correspond to rigid-body rotations of approximately 1.5 degrees, which explains the strong similarity between the isolator base and statue displacement time histories. Therefore it is possible to state that the proposed analytical model allows to define a set of structural parameters able to correctly simulate the dynamic of the structure under investigation.

Fig. 3. Displacements of the KSJ isolator at point D1 vs time: experimental results (red line), optimized results by using the PSO algorithm: (a) x direction; (b) y direction.

Fig. 4. Displacements of the statue at point D2 with respect to the base isolation vs time: experimental results (red line), optimized results by using the PSO algorithm: (a) x direction; (b) y direction.

6 Conclusions

A novel bi-directional base-isolation device, based on the patented Kinematic Steel Joint (KSJ) system and previously developed and validated in earlier studies [20, 23], has been selected as the case study for the development and verification of an analytical model aimed at protecting statue-like structures from seismic excitations. A combined experimental and analytical investigation, using a full-scale replica of the Goddess of Morgantina statue, has been conducted to assess the capability of the proposed model in reproducing the dynamic behavior of the isolated system.

The analytical model incorporates restoring forces due to the curvature of the isolator, frictional forces under both sticking and sliding conditions, and the inertial effects of the statue. Rotational displacements of the statue relative to the isolated base have been represented through a rotational stiffness of connection, while overturning was prevented by the rigid connection between the statue and isolator. Model calibration has been achieved through a Particle Swarm

Optimization procedure that minimized the RMS error between experimental and numerical displacements at both the isolator base and the statue.

Results show very good agreement between the experimental data and the optimized numerical model, particularly during the most intense portions of the motion. The statue exhibits small rigid-body rotations of approximately 1.5° , explaining the close similarity between the isolator base and statue displacement time histories.

Overall, the study demonstrates that the proposed analytical model accurately reproduces the dynamic behavior of the protected structure.

References

1. Augusti, G., Ciampoli, M., Airoidi, L.: Mitigation of seismic risk for museum contents: An introductory investigation. In: 10th World Conference on Earthquake Engineering, Vol. 9, pp. 5995–6000. Balkema, Rotterdam, the Netherlands (1992).
2. Housner, G.W.: The behavior of inverted pendulum structures during earthquakes. *Bulletin of the Seismological Society of America* **53**(2), 403–417 (1963).
3. Aslam, M., Godden, W.G., Scalise, D.T.: Earthquake rocking response of rigid bodies. Tech. Rep. EERC 2003-07, Prepared for the Department of Energy under Contract W-7405-ENG-48 (1978).
4. Ishiyama, Y.: Motions of rigid bodies and criteria for overturning by earthquake excitations. *Earthquake Eng. Struct. Dyn.* **10**(5), 635–650 (1982).
5. Calì, I., Marletta, M.: Passive control of the seismic rocking response of art objects. *Eng. Struct.* **25**(8), 1009–1018 (2003).
6. Di Egidio, A., Contento, A.: Base isolation of slide-rocking non-symmetric rigid blocks under impulsive and seismic excitations. *Eng. Struct.* **31**(11), 2723–2734 (2009).
7. Koumousis, V., Moysidis, A.: On the dynamic behavior of nonlinear lightweight isolator for museum artifacts. *Soil Dyn. Earthq. Eng.* **117**, 251–262 (2019).
8. Berto, T.F., Saetta, A.: Seismic risk mitigation technique for art objects: Experimental evaluation and numerical modelling of double concave curved surface sliders. *Bull. Earthq. Eng.* **11**(5), 1817–1840 (2013).
9. Roussis, P., Odysseos, S.: Slide-rocking response of seismically-isolated rigid structures subjected to horizontal ground excitation. In: Proc. 2nd European Conference on Earthquake Engineering and Seismology, Istanbul, Turkey (2014).
10. Kavvadias, I., Vasiliadis, L., Elenas, A., Koutsoupakis, K.: Fragility assessment of base isolated free standing museum artifacts. In: Proc. 6th Conference on Computational Methods in Structural Dynamics and Earthquake Engineering, Limenas Chersonisou, Greece (2019).
11. Vestroni, F., Di Cinto, S.: Base isolation for seismic protection of statues. In: Proc. 12th World Conference on Earthquake Engineering, Auckland, New Zealand, 30 January–4 February 2000. New Zealand Society for Earthquake Engineering (2000).
12. Pellecchia, D., Lo Feudo, S., Vaiana, N., Dion, J.-L., Rosati, L.: A procedure to model and design elastomeric-based isolation systems for the seismic protection of rocking art objects. *Comput.-Aided Civ. Inf. Eng.* **37**(10), 1298–1315 (2021).
13. Neurohr, T., McClure, G.: Shake table testing of museum display cases. *Can. J. Civ. Eng.* **35**(12), 1353–1364 (2008).

14. Prota, A., Zito, M., D'Angela, D., Toscano, G., Ceraldi, C., Fiorillo, A., Magliulo, G., Perrone, D.: Preliminary results of shake table tests of a typical museum display case containing an art object. *Adv. Civ. Eng.* **2022**(1) (2022).
15. Aoki, T., Kawaguchi, K.: Shake-table test on statue pedestals with sliding isolators. In: *Seismic Mitigation for Museum Collection. Papers from J. P. Getty Museum of Western Art Jointly Sponsored Symposium*, Los Angeles, California, USA (2011).
16. Zou, X., Yang, W., Liu, P., Wang, M.: Shaking table tests and numerical study of a sliding isolation bearing for the seismic protection of museum artifacts. *Journal of Building Engineering* **65**, 105725 (2023).
17. Cocuzza Avellino, G., Cannizzaro, F., Di Martino, A., Valenti, R., Paternò, E., Calì, I., Impollonia, N.: Numerical and experimental response of free-standing art objects subjected to ground motion. *International Journal of Architectural Heritage* **16**(11), 1666–1682 (2021).
18. Huang, B.F., Günay, S., Lu, W.S.: Seismic assessment of freestanding ceramic vase with shaking table testing and performance-based earthquake engineering. *Journal of Earthquake Engineering* **26**(15), 7956–7978 (2021).
19. Berto, L., Di Sarno, L., Rocca, E.: Double-concave curved-surface slider isolators for museum artifacts. *Engineering Structures* **300**, 115458 (2025).
20. Oliva, M., Lo Iacono, F., Navarra, G.: Seismic performance assessment of a novel 2-D base isolation device for protecting statues via shaking table tests. The case study of the Statue of Goddess of Morgantina. *Int. J. Archit. Herit.* **1–19** (2025).
21. Guerrini, G., Ausenda, G., Graziotti, F., Penna, A.: An innovative seismic isolation device based on multiple articulated quadrilateral mechanisms: Analytical study and shake-table test. In: *Proc. of XVIII ANIDIS Conference*, Ascoli Piceno, Italy, 15–19 September 2019
22. Lo Iacono, F., Navarra, G., Oliva, M., Cascone, D.: Experimental investigation of the dynamic performances of the L.E.D.A. shaking tables system. In: *Proc. of the 23rd AIMETA Conference 2017*, Salerno (Italy), vol. 5, pp. 897–915 (2017).
23. Alberti, E., Lo Iacono, F., Navarra, G.: Shake-Table Test Assessment of a Base-Isolation Device for the Seismic Protection of the Goddess of Morgantina Statue. *Materials Research Proceedings* **26**, Materials Research Forum LLC (2023).
24. Alaimo, R., Giarrusso, R., Montana, G., Quinn, P.: Petrographic and micropalaeontological data in support of a Sicilian origin for the statue of Aphrodite. In: *Cult Statue of a Goddess, Summary of Proceedings from a Workshop*, 1–10. Getty Publications, Los Angeles (2007).
25. Marconi, C.: L'identificazione della Dea di Morgantina. *Prospettiva: Rivista di Storia dell'Arte Antica e Moderna* **2011**(141), 2–15 (2011).
26. Liuzzo, M., Caraccio, D., Floriano, L.: Transizioni digitali e fisiche per i beni museali / Digital and physical transitions for museum asset. In: Cannella, M., Garozzo, A., Morena, S. (eds.) *Transizioni. Atti del 44° Convegno Internazionale dei Docenti delle Discipline della Rappresentazione / Transitions. Proceedings of the 44th International Conference of Representation Disciplines Teachers*, 2894–2913. Milano: FrancoAngeli (2023).
27. L.Y. Lu, M.H. Shih, C.Y. Wu: Near fault seismic isolation using sliding bearings with variable curvatures. In: *Proc. of 13th World Conference in Earthquake Engineering*, paper no. 3264, Vancouver, Canada (2004).
28. A. Mokha, M.C. Constantinou, A.M. Reinhorn, V.A. Zayas: Experimental study on Friction-Pendulum Isolation System. *Journal of Structural Engineering* **117**, 1201–1217 (1991).
29. Arias, A.: A measure of earthquake intensity. *Seismic Design for Nuclear Power Plants* (R.J. Hansen ed.), M.I.T. Press, 438–483 (1970).